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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

AM2102-1 – OPERATING SYSTEM ESSENTIALS
(AI&ML,CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which one is not true with respect to operating system?
A) System software which acts as an intermediary between user and hardware.
B) Keeps the complicated details of the hardware hidden from the user.
C) Provides user with easy and simple interface
D) Virtualization is core of OS
2. Second generation of OS is
A) Transistors and Batch Systems
B) Vacuum tube
C) ICs
D) Multiprogramming
3. Memory hierarchy order based on increasing speed is
A) Disk, Main memory, cache, registers
B) Registers, cache, main memory, disk
C) Disk, Primary memory, Temp registers, cache
D) none
4. Process of loading the kernel is known as
A) boating
B) booting
C) BIOS starting
D) init
5. Process memory is divided into ---- sections
A) 3
B) 2
C) 4
D) 5
6. Thread consists of ----
A) Two PCs, 64 registers, two stacks
B) a PC, set of registers, a stack
C) Program, registers, queue
D) A PC, a stack, a queue
7. Work of short term scheduler is ---
A) To schedule the CPU to a process
B) To schedule the memory to a process
C) To select process from disk
D) all
8. Dispatch latency is---
A) The time taken to stop one process and start another
B) Time taken from the start of a process execution till completion
C) Delay incurred in dispatching the process
D) Time taken by process to wait in ready queue
9. If Resource allocation graph contains cycle then
A) 100% chance of deadlock
B) 50% chance of deadlock
C) Deadlock happens only if one instance of resource is available
D) Deadlock happens only if many instances of resources are available
10. Banker's algorithm is a deadlock ----
A) Detection algorithm
B) Recovery algorithm
C) Both A and B
D) Avoidance algorithm
11. Resource allocation Graph (RAG) can be used for deadlock avoidance only when-----of resources are available.
A) Multiple instances
B) Single instances
C) both
D) Two instances
12. The circular wait condition can be prevented by
A) using thread
B) using pipes
C) defining a linear ordering of resource types
D) All of mentioned

13. _____ is the concept in which a process is copied into the main memory from the secondary memory according to the requirement.
- A) Demand paging B) Paging
C) Segmentation D) Swapping
14. When a program tries to access a page that is mapped in address space but not loaded in physical memory, then _____
- A) segmentation fault occurs B) fatal error occurs
C) page fault occurs D) no error occurs
15. A process is thrashing if _____
- A) it is spending more time paging than executing B) page fault occurs
C) it is spending less time paging than executing D) No swapping takes place
16. Binding of instructions and data to memory addresses can be done at _____
- A) Compile time B) Load time
C) Execution time D) All of the mentioned
17. What is the mounting of file system?
- A) creating of a filesystem B) deleting of a filesystem
C) attaching portion of the file system into a directory structure D) removing the portion of the file system into a directory structure
18. When will file system fragmentation occur?
- A) used space is not contiguous B) multiple files are non-contiguous
C) unused space is non-contiguous D) unused space or single file are not contiguous
19. What is the primary purpose of a hypervisor in virtualization?
- A) To create virtual hard disks B) To allocate hardware resources and manage virtual machines
C) To manage the network connections D) To manage the guest operating systems
20. What is a snapshot in virtualization?
- A) A screenshot of the virtual machine's desktop B) A copy of the virtual machine's entire hard disk
C) A point-in-time image of the virtual machine's state D) A backup of the host operating system

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I				Marks	BT*	CO*	PO*	
1.	a)	Discuss types of system calls with example.		6	L1	1	1	
	b)	Discuss different types of thread models.		10	L2	2	1	
2.	a)	Differentiate between monolithic and microkernel operating systems.		6	L2	1	1	
	b)	Show how different scheduling criteria will optimize the scheduling.		5	L3	2	1	
	c)	Solve the following scenario for Priority scheduling, round robin scheduling (time slice:4ms) and calculate the turnaround and waiting times.						
		Process id	CPU Burst time	Priority				
		P1	10	3				
		P2	8	1				
		P3	12	2	5	L3	2	1
3.	a)	Explain the process state diagram.		6	L1	2	1	
	b)	Illustrate FCFS and shortest job first scheduling for the given scenario and calculate the turnaround and waiting times.						
		Process id	CPU Burst time	Arrival time				
		P1	12	4				
		P2	3	1				
		P3	6	7				
		P4	8	10	6	L3	2	1
	c)	Compare and contrast Process and Thread.		4	L2	2	1	

Unit – II

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|-------|--|---|----|---|---|
| 4. a) | Discuss the various conditions that are to be characterized and satisfied for the deadlock to occur. | 4 | L2 | 3 | 1 |
| b) | Show with an example how the page address translation takes place? | 6 | L2 | 4 | 1 |
| c) | Given the following reference string, show how Optimal and Least recently used (LRU) page replacement algorithm works? Give the number of page misses and page hits. Uses 4 frames:
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 1, 5, 6, 7, 2, 3. | 6 | L3 | 4 | 1 |
| 5. a) | Discuss the steps in handling page fault with a diagram. | 4 | L1 | 4 | 1 |
| b) | Show how Resource allocation graph (RAG) helps in detection of deadlocks? | 6 | L2 | 3 | 1 |
| c) | Given the following reference string, show how First In First Out (FIFO) and Least recently used (LRU) page replacement algorithm works? Give the number of page misses and page hits. Uses 3 frames:
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 1, 6, 5. | 6 | L3 | 4 | 1 |
| 6. a) | With figure show how limit and relocation registers are used in effective memory protection. | 4 | L2 | 4 | 1 |
| b) | Explain with diagram how the paging with TLB helps in reduction of time in memory access? | 6 | L2 | 4 | 1 |
| c) | Write short notes on i) demand paging steps ii) Copy on Write (COW). | 6 | L1 | 4 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Discuss with diagrams any 4 logical structures of directories. | 8 | L1 | 5 | 1 |
| b) | How the type1 and type 2 hypervisors are used in virtualization? | 4 | L2 | 5 | 1 |
| c) | Differentiate virtualization and paravirtualization? | 4 | L2 | 5 | 1 |
| 8. a) | Discuss with neat diagrams the directory allocation methods. | 9 | L2 | 5 | 1 |
| b) | Which are the 5 essential characteristics listed by National institute of standards and technology for clouds. | 5 | L2 | 5 | 1 |
| c) | Which are the two data structures used for file directory implementation. | 2 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
